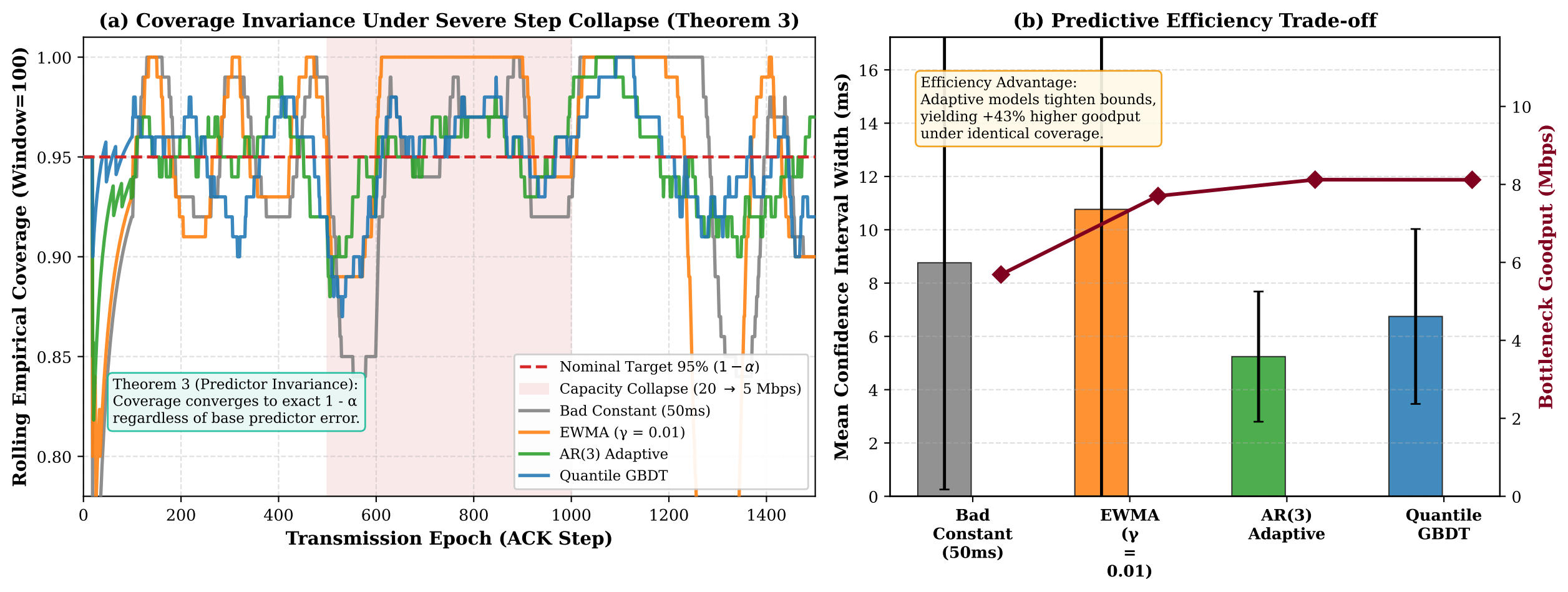
(a) Coverage Invariance Under Severe Step Collapse (Theorem 3)
Rolling Empirical Coverage (Window=100)
Transmission Epoch (ACK Step)
Theorem 3 (Predictor Invariance):
Coverage converges to exact 1 - α
regardless of base predictor error.
Nominal Target 95% (1 − α)
Capacity Collapse (20 → 5 Mbps)
Bad Constant (50ms)
EWMA (γ = 0.01)
AR(3) Adaptive
Quantile GBDT
(b) Predictive Efficiency Trade-off
Mean Confidence Interval Width (ms)
Bottleneck Goodput (Mbps)
Efficiency Advantage:
Adaptive models tighten bounds,
yielding +43% higher goodput
under identical coverage.
Bad Constant (50ms)
EWMA (γ = 0.01)
AR(3) Adaptive
Quantile GBDT